

WELCOME!!

ASSIGNMENT NO:1

TOPIC : NESTED FOR LOOP WITH EXAMPLE

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Introduction

- A loop is used to repeat instructions.
- A **nested for loop** is a loop inside another loop.
- Used for:
 - Pattern printing
 - Working with rows & columns
 - 2D data structures

Engineering Knowledge (Concept)

- Outer loop → controls rows
- Inner loop → controls columns

Syntax:

```
for i in range(n):  
    for j in range(m):  
        print(i, j)
```

Inner loop executes fully for each outer loop iteration

⌚ TIME COMPLEXITY:

If both loops run n times

👉 Total complexity = $O(n^2)$

Problem Analysis

Problem: Print a square star pattern

Output:

```
* * *  
* * *  
* * *
```

- Need 3 rows and 3 columns
- Requires repeating printing in rows and columns
- Solution: Use nested loops

Design of Solution

Algorithm:

- Start
- Set number of rows
- Use outer loop for rows
- Use inner loop for columns
- Print "*" .
- Move to next line
- End

⚠ Common Mistakes:

- Wrong indentation
- Incorrect loop range
- Missing newline
- Infinite loops

Implementation & Complex Problem Solving

Square Pattern Code:

```
rows = 3

for i in range(rows):
    for j in range(rows):
        print("*", end=" ")
    print()
```

Output:

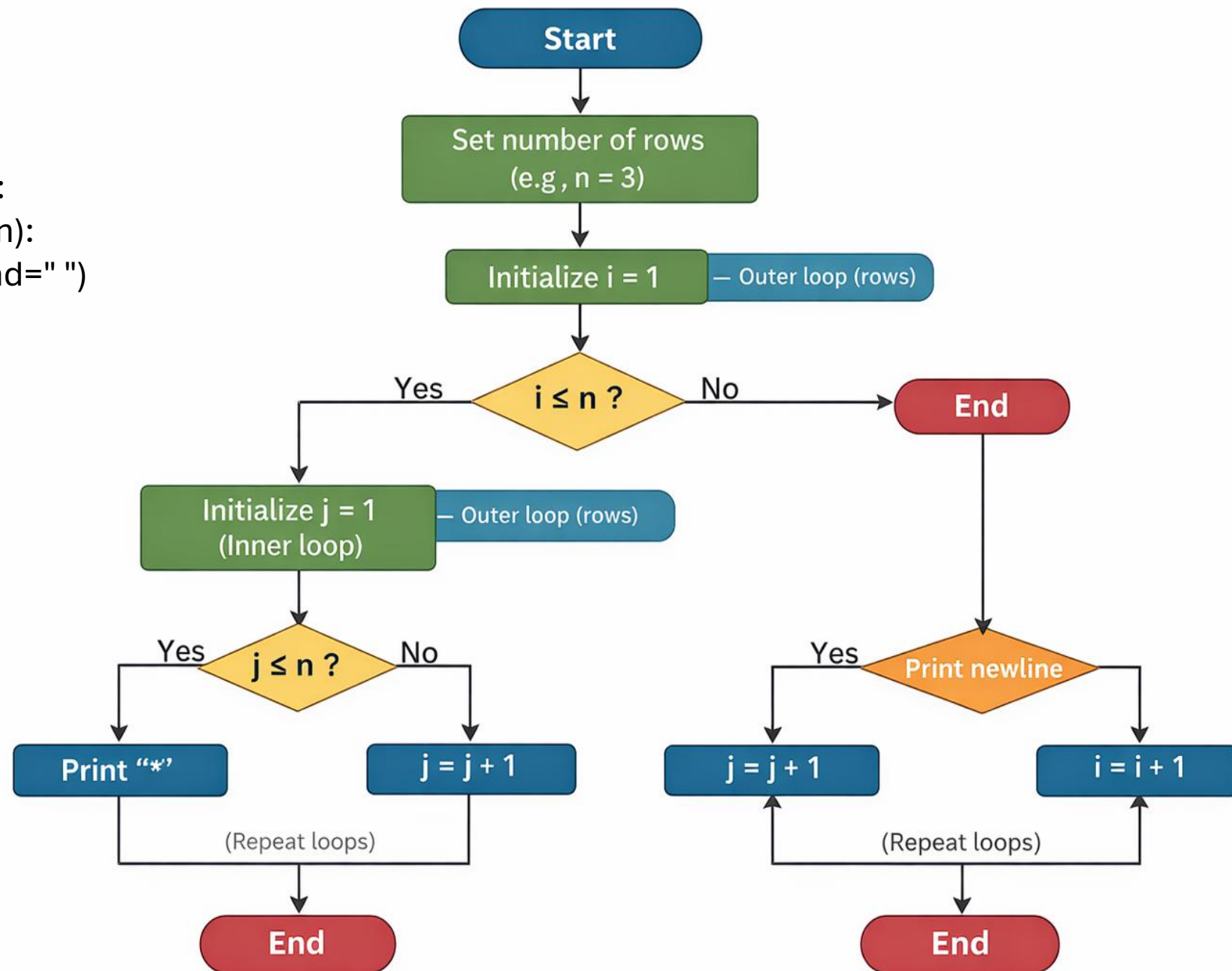
```
* * *
* * *
* * *
```

Triangle Pattern code :

```
for i in range(1, 4):
    for j in range(i):
        print("*", end=" ")
    print()
```

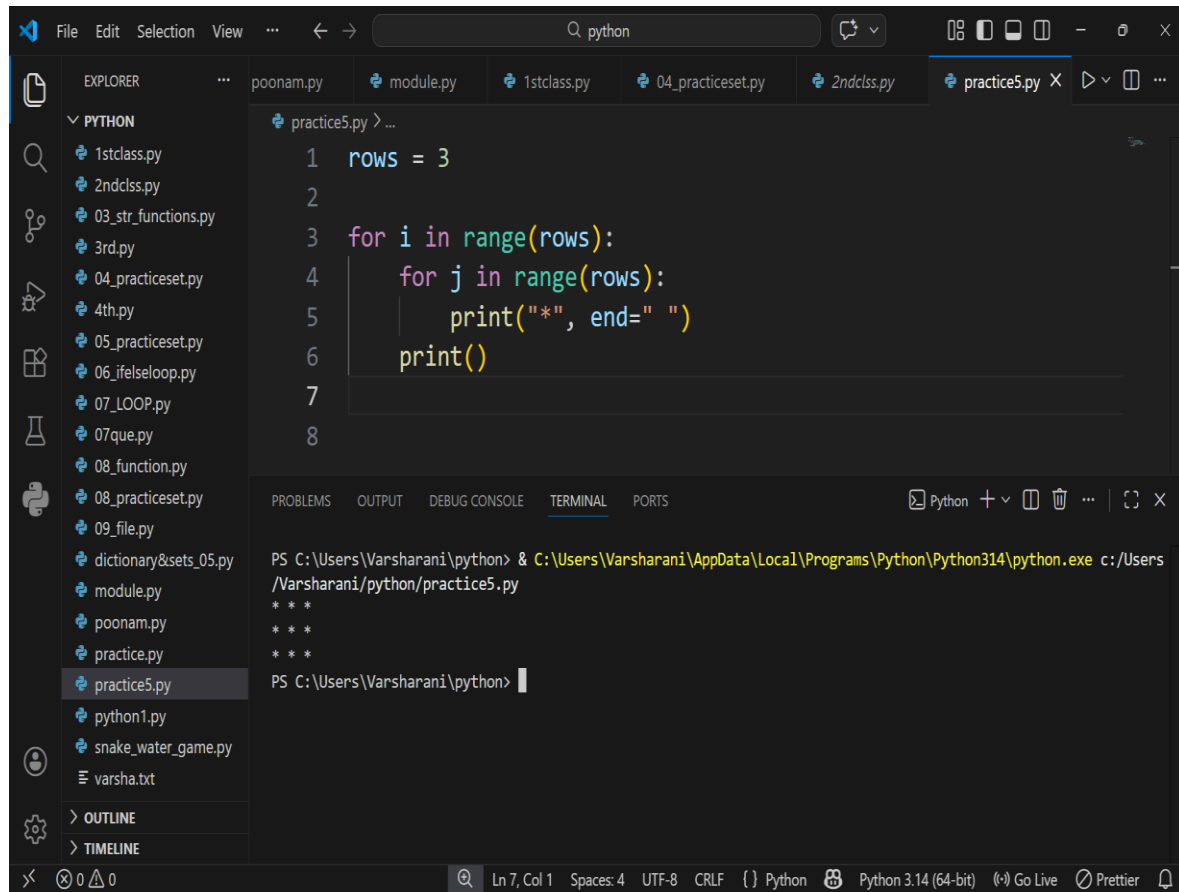
```
*
* *
* * *
```

```
n = 3
i = 1
for i in range(n):
    for j in range(n):
        print("*", end=" ")
    print()
```



VS CODE

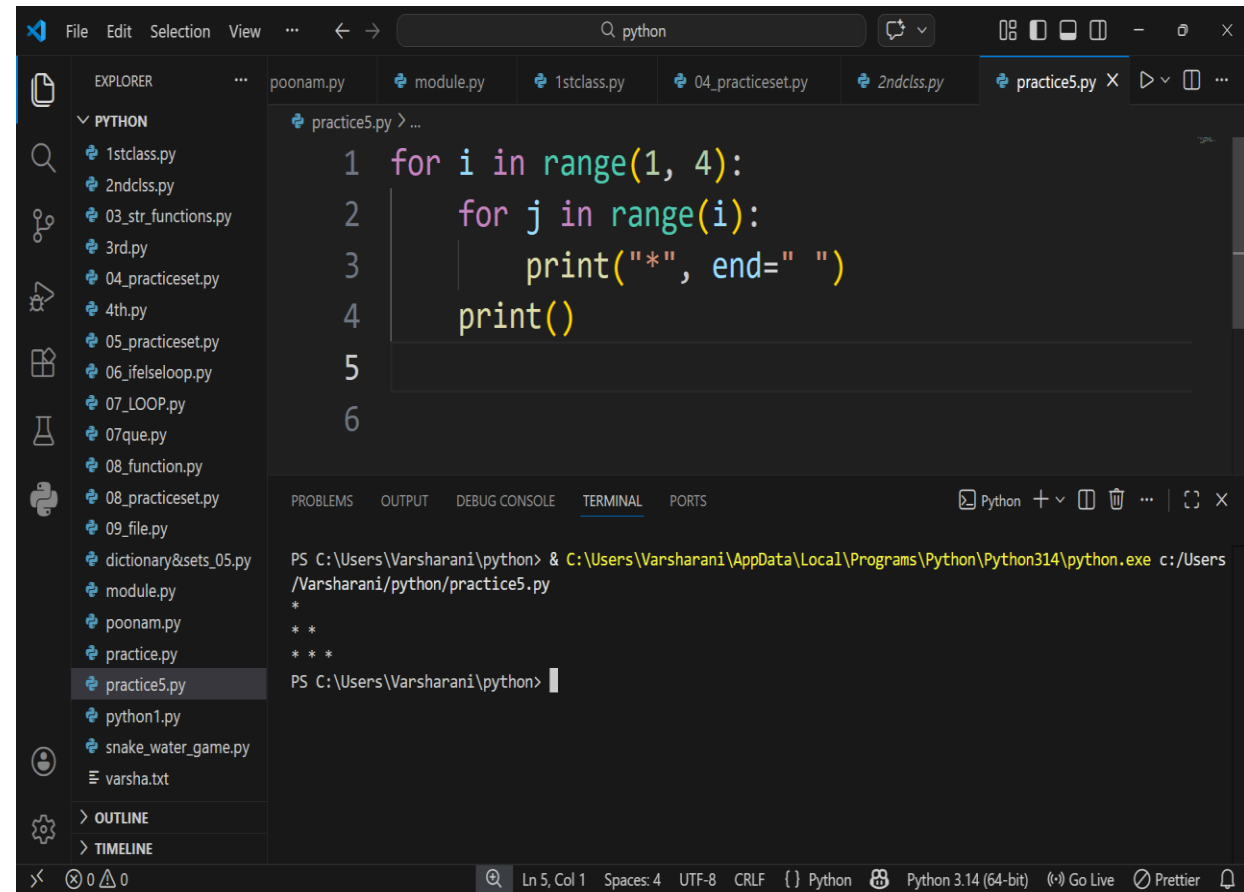
Square Pattern Code



```
1 rows = 3
2
3 for i in range(rows):
4     for j in range(rows):
5         print("*", end=" ")
6     print()
7
8
```

```
PS C:\Users\Varsharani\python> & C:\Users\Varsharani\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Varsharani/python/practice5.py
* * *
* * *
* * *
```

Triangle Pattern code



```
1 for i in range(1, 4):
2     for j in range(i):
3         print("*", end=" ")
4     print()
5
6
```

```
PS C:\Users\Varsharani\python> & C:\Users\Varsharani\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Varsharani/python/practice5.py
*
* *
* * *
```


Advantages:

1. Easy for pattern problems
2. Works with 2D data
3. Improves logic building

Conclusion:

- Nested loops help solve complex problems
- Widely used in real-world programming

THANK

YOU!!